Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/22/2006
PI Name:	Hasenstein, Karl Ph.D.		
Project Title:	Spatial and Temporal Analysis of Gravitropism In Chara Rhizoids		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hasenstein@louisiana.edu	**Fax:**	FY 337-482-5834
PI Organization Type:	UNIVERSITY	**Phone:**	337-482-6750
Organization Name:	University of Louisiana - Lafayette		
PI Address 1:	PO Box 42451		
PI Address 2:	300 E. St. Mary Boulevard		
PI Web Page:	http://www.ucs.louisiana.edu/		
City:	Lafayette	**State:**	LA
Zip Code:	70504-2451	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	08/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 6/30/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNA04CK48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposal focuses on the sequence of events of gravisensing in rhizoids of the green alga Chara as a model system for higher plants. The goal is to analyze the sequential and spatial coordination of cellular responses to reorientation. Despite the long-standing correlation between the sedimentation of statoliths and subsequent growth responses, little is known about the transduction of the stimulus, how the positions of the putative sensors are maintained or the steps that lead to differential elongation of either the cell wall of (Chara) rhizoids or an entire cross section of a root or stem. Based on a previously developed microinjection system capable of delivering micron-sized particles (comparable to endogenous statoliths) we propose to use these particles as statoliths and in their oligo(dT)-coated version as tools for extraction of mRNA and subsequent RT-PCR analyses. The magnetic properties of the beads allow their positioning to regions of interest and facilitate the recovery of particles after binding of mRNA. Together with an external magnet the magnetic properties of the beads can also be used to vary the forces acting on artificial statoliths, comparable to gravistimulation with varying forces. Based on this bead-based system and control experiments that will probe various areas of the rhizoid with oligo(dT)-coated glass probes (= solid phase gene extraction, SPGE), we will analyze the distribution and quantity of mRNA's in different zones of the cell during the graviresponse cycle. The proposed work will identify genes and their localization in a single graviresponding cell. The work will result in a novel, stimulus-response analysis that will correlate temporal and spatial information with gene expression patterns during graviresponse. The proposed approach will also discern between primary (= graviresponse) messages and induced secondary (e.g., stress) signals. The experiments will deepen our understanding of gravity effects on a molecular level that is focused on the gravisensing and transduction process. The aim of this project is to enhance our understanding of the nature of perceptions and responses of plants to changes in their environment, especially gravity. The proposed work is aimed at providing fundamental knowledge on plant growth, perception of forces and the sequential and spatial organization of responses to such changes. The project focuses on the effects of gravity, but also includes the perception of external (friction, light, mechanical stress) and internal (changes in viscosity, cellular perturbations) factors which are vital for the development and growth of plants. The project is in line with NASA's quest to improve the quality of life here and extend it to space exploration. It is essential to achieving long-duration human presence in and exploration of space. The proposed work combines novel cellular and molecular approaches to understanding of the perception of and response to the gravity signal and subsequent transduction. The genetic analyses represent not only a new technology but also the first attempt to characterize a valuable and widely studied model system for which almost no genetic data are available. The proposed approach will establish a thorough and, at least for plant systems, novel screening methodology. Specific experiments are designed to follow gene expression over time and (cellular) space. The research will integrate the training of post-docs with molecular, general plant biology and specifically space biology. The project includes many technical aspects (microinjection, genetic analyses, growth studies, hormonal analyses, and microscopy) as well as numerous conceptual facets (spatial and temporal analysis of events and distributions, modeling of gene expression and identification of factors and processes contributing to differential elongation). The potential payoff is high in identifying key regulatory elements for the interaction of physical forces (gravity), auxin and genetic integration and elucidating how these factors are controlled and contribute to the response of plants to gravity. The project will train plant biologists in gravitational biology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Aside from the research on gene activation, sequestration and transport, the current work breaks new ground in rapid (< minute) mRNA extraction. Potential future applications may eliminate or reduce the need for destructive tissue biopsies for e.g., cancer or other disease diagnosis.
Task Progress:	Solid Phase Gene Extraction (SPGE) depends on poly(T)-ologonucleotides that are chemically linked to the surface of glass (or other) material. The glass rod is beveled to <10 µm to be easily inserted into cells or tissue and to hybridize with the poly(A)tail of mRNA. The functionality of the SPGE technique was further tested in vivo by determining the expression levels of three genes in Arabidopsis seedlings. We choose the low abundant IAA5 gene that is induced by the external application of the plant hormone auxin (indole-3-acetic acid, IAA). These data were normalized to the actin encoding genes ACT2 and ACT8. We standardized the coating and extraction procedure (data not shown) and because of the high fidelity are now confident that SPGE-based mRNA extractions do not require housekeeping gene analysis. Analysis of the distribution of nanos and bicoid mRNA in Drosophila eggs validated SPGE as a reliable mRNA extraction method. Importantly, when reverse transcription was omitted (-RT controls) no amplification was detected in real-time PCR assays, indicating the absence of genomic DNA.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Scherp P, Hasenstein KH. "Viscous anisotropy of cytoplasm facilitates lateral displacement of statoliths in Chara rhizoids. " ASGSB meeting, Reno NV, November 2005. Gravit Space Biol. 2005 Nov;19(1):50. , Nov-2005